Qualcomm Snapdragon 8155 Automotive Chip

An in-vehicle smart cockpit chip introduced by Qualcomm Inc.



Qualcomm Snapdragon 8155 Automotive Chip (full name: SA8155P) is an automotive smart cockpit chip launched by Qualcomm Inc., released in 2019, and belongs to Qualcomm's third-generation Snapdragon Automotive Digital Cockpit Platform. The chip is improved from Snapdragon 855 , adopts 7-nanometer process technology, with AI computing power reaching 8 TOPS, mainly used in the field of smart cockpits. [2]

The chip's CPU adopts an octa-core "1+3+4" Tri-Cluster Design, including 1 big core customized based on ARM Cortex-A76 with a maximum clock frequency of 2.96GHz, 3 performance cores with a clock frequency of 2.42GHz, and 4 efficiency cores with a clock frequency of 1.8GHz. The GPU is Adreno 640, and its Neural Processing Unit (NPU) can run at up to 900MHz. The SA8155P chip natively supports Hypervisor virtualization technology, enabling "One Chip, Multiple Systems". It can simultaneously support up to 6 camera inputs and can drive up to 4 2K-resolution screens or 3 4K-resolution screens. In terms of network connectivity, it supports 5G, Wi-Fi 6 , and Bluetooth 5.0, with support for up to 12GB of memory. [2]

The chip has been applied to multiple vehicle models, including some models of Chinese independent brands such as Great Wall Motors, GAC Group, Geely, Chery, BYD, and Jetour. Its global cumulative shipments have exceeded 2 million units. In 2021, the WEY Mocha was equipped with this chip. The Li Auto L9 adopted a dual-Snapdragon-8155-chip solution. Some automakers provide hardware upgrade services for this chip to owners of already-sold vehicle models. In March 2026, the Toyota Corolla model sparked controversy because the Qualcomm Snapdragon 8155P chip it was advertised as being equipped with was actually downclocked and had its storage chips replaced. Its successor product is the Snapdragon 8295 (SA8295P) chip, which adopts a 5nm process. [10]

Chinese name	高通骁龙8155车载芯片	Product type	Automotive smart cockpit chip
Launch date	2019		

Development History

The Qualcomm Snapdragon 8155 Automotive Chip (full name SA8155P) is the core chip of the third-generation Snapdragon Automotive Digital Cockpit Platform released by Qualcomm Inc. in 2019. Derived from Qualcomm's flagship smartphone chip, the Snapdragon 855 , it is the industry's first Automotive-grade digital cockpit SoC (System on Chip) to adopt the 7-nanometer process technology. [1]

In 2021, the WEY Mocha became the world's first vehicle model to be equipped with this chip, followed by models such as the WM Motor W6 and the Geely Xingyue L. [26]

Compared to its predecessor, the Snapdragon 820A, which used a 14-nanometer process, the Snapdragon 8155 improved the process from 14-nanometer to 7-nanometer, increased the number of cores from 4 to 8, boosted CPU computing power by 177%, doubled the memory bandwidth, and increased the number of supported in-vehicle screens from 3 to 6. In 2021, the Qualcomm Snapdragon 8155 chip was widely adopted in China's new energy vehicle intelligent cockpits. Although its successor, the Snapdragon 8295 chip, has been released, as of 2023, the four-year-old Snapdragon 8155 chip remains widely used among automotive chips. [3]

On August 6, 2026, the Chery Jetour Zongheng F700 will be launched. The vehicle is expected to be equipped with the mainstream 8155 chip or a higher-specification computing platform to ensure system smoothness and functional expandability. [45]

On August 20, 2026, smart China officially announced that the smart #3 Sport Racing Edition would make its debut at the Chengdu Auto Show, with the new car equipped with the Qualcomm Snapdragon 8155 chip. On September 1, the 2027 Exeed Lanyue was officially launched, featuring the Qualcomm Snapdragon 8155 chip and the Lion 5.0 Smart Cloud System in its in-car infotainment, compatible with multiple mainstream mobile phone connectivity options. On September 9, the all-new Jetour X90 was officially launched, and its chip was upgraded to the Qualcomm Snapdragon 8155. While not top-tier in 2026, it was already a class-above configuration for a fuel SUV in the 100,000-yuan price segment. [47]

Product Design

The Qualcomm Snapdragon 8155 (full name SA8155P) is the flagship product of Qualcomm's third-generation Snapdragon Automotive Digital Cockpit Platform. It is optimized from the smartphone-oriented Snapdragon 855 chip to meet automotive-grade requirements. It is the world's first automotive-grade digital cockpit SoC manufactured using a 7-nanometer process, designed to provide a high-performance, high-reliability computing platform for intelligent cockpits. [24]

The chip is manufactured using TSMC's 7-nanometer process technology. Its CPU employs an octa-core design based on a "1+3+4" tri-cluster architecture. This includes 1 high-performance Gold core (clocked at 2.84GHz or 2.96GHz) customized from ARM Cortex-A76, 3 performance Gold cores (clocked at 2.42GHz) customized from ARM Cortex-A76, and 4 efficiency Silver cores (clocked at 1.8GHz) customized from ARM Cortex-A55. The CPU computing power is approximately 95K to 105K DMIPS. [27]

The GPU is the Adreno 640, with an operating frequency of 700MHz, delivering approximately 1000 to 1100 GFLOPS of computing power. The chip integrates a dedicated Neural Processing Unit (NPU) that can operate at frequencies up to 900MHz, with a typical AI computing power of 8 TOPS. [27]

In terms of multimedia and display, the chip can support up to 6 camera inputs simultaneously and drive up to 4 screens with 2K (2560x1440) resolution or 3 screens with 4K (4096×2160) resolution, supporting multi-screen interaction. [2]

Regarding connectivity features, it integrates next-generation wireless connectivity technologies, supporting 5G cellular networks, Wi-Fi 6, and Bluetooth 5.0. Its Bluetooth 5.0 bandwidth reaches 2Mbps, offering significant improvements in effective transmission range and power consumption compared to previous-generation products. [2]

In terms of physical specifications, the chip uses FCBGA+HS packaging measuring 23mm × 23mm × 2.26mm, with a BGA pitch of 0.7mm. It complies with the AEC-Q100 Grade 3 automotive standard, with an operating junction temperature range of -40°C to +105°C. [41]

The SA8155P chip natively supports Hypervisor virtualization technology (such as QNX Hypervisor), enabling "one-chip, multi-system" capability. For example, it can run the QNX safety system and the Android infotainment system simultaneously to meet the dual requirements of functional safety and a rich application ecosystem for intelligent cockpits. [22]

Through the advanced 7-nanometer process technology, heterogeneous computing architecture, and the integration of technologies such as high-speed connectivity, multi-display support, and virtualization, the Snapdragon 8155 is designed to deliver a powerful computing performance, low power consumption, and highly reliable solution for automotive intelligent cockpits. [2]

Product Models and Platform

The full name of the Qualcomm Snapdragon 8155 Automotive Chip is the Qualcomm Snapdragon SA8155P (or SA8155P). It is the flagship (Flagship) product within Qualcomm's third-generation Snapdragon Automotive Digital Cockpit Platform (also known as the third-generation intelligent cockpit platform or the third-generation digital cockpit platform). Other products in the same generation platform include the performance-level (Performance) SA6155P and the premiere-level (Premiere) SA8195P. [25]

Market Applications and Performance

Nearly one hundred vehicle models have been equipped with the Qualcomm Snapdragon 8155 chip. In 2023, intelligent cockpit domain controllers based on the SA8155P captured approximately 30% of the market share for 8155 cockpit

domain controllers. As of June 2025, its global cumulative shipments have exceeded 2 million units. The price range of vehicles equipped with this chip spans multiple market segments from economy to luxury. In the 100,000-yuan price segment, several models feature it as standard equipment across all trims, and there are over 10 models with starting prices below 200,000 yuan that are equipped with this chip. [1]

This chip has been used in more than 20 models and over 80 vehicle model programs of mainstream Chinese independent brands such as Great Wall Motors, GAC Group, Geely, Chery , BYD, and Jetour . Models equipped with this chip include the Geely Xingyue L and Lynk & Co 09, among others. In August 2026, the Xingyue L i-HEV Smart Hybrid Edition made this chip standard across the entire lineup. Among these models, the WEY Mocha was the world's first to feature the chip; the Li Auto L9 adopted a dual-Snapdragon 8155 chip solution to meet multi-screen requirements; and the BMW New 3 Series (2025 model) has also been equipped with it across the entire lineup. Models from traditional independent brands, such as the Bestune T90, Chery Explore 06, Chery QQ3 EV, Changan UNI-T 2023 model, GAC Trumpchi Yingku, Roewe RX5 MAX, Maxus G90, Fifth-generation Tiggo 8 flagship edition, and Great Wall Motorcycle 2024 Show Model, are all equipped with this chip. The Nano 01 Zhiqu Edition (launched in July 2026) also features this chip; its high computing power enables multiple tasks such as navigation, audio/video, and vehicle control to run simultaneously, and it supports intelligent functions such as the Doubao and DeepSeek dual-AI voice large models and the Tianyuan OS 2.0 system. Among new forces and new energy vehicle brands, the Leapmotor C11 and WM Motor W6 are equipped with the chip, and the new Neta U Pro will also be equipped with it. Among joint-venture and foreign brands, the 2024 Toyota Camry mid-to-high trim models, Toyota bZ4X, and Mercedes-Benz smart #1 have already adopted the chip, while the Ford Mustang Mach-E can provide users with hardware upgrade services. On September 21, 2026, the FAW Toyota RAV4 Adventure Edition was officially launched. Based on the 2026 Dual-Engine 2.0L Two-Wheel Drive Elite Edition, the vehicle has the Qualcomm Snapdragon 8155 (SA8155P) chip built into its cockpit and is equipped with a 15.6-inch floating center console screen, a full LCD instrument cluster, and a HUD Head-Up Display, supporting three-screen interaction. On September 22, 2026, the all-new QQ3 Modern Edition was officially launched, equipped with the Snapdragon 8155 chip. [1]

This chip is often used as a reference for evaluating smart cockpit performance. Some automakers provide owners of already-sold models with hardware upgrade services for this chip, and brands such as Li Auto have also announced paid or free Snapdragon 8155 cockpit upgrade services for existing owners. [14]

Tests on models equipped with this chip show that navigation startup time is reduced to 2.3 seconds, app switching response speed is improved by 40%, and voice command recognition accuracy reaches over 92%. [12]

Technical Features and Advantages

The Qualcomm Snapdragon 8155 chip (full name SA8155P) is manufactured using the advanced 7-nanometer process technology, making it the industry's first automotive-grade digital cockpit solution to adopt this process. For automotive-grade chips, a more advanced process technology not only signifies performance improvements but also meets the stringent requirements for stability throughout the vehicle's long lifecycle. Its performance is three times that of the original Qualcomm 820 platform, and its memory bandwidth has also doubled. Compared to the 820A, the 8155P integrates the WiFi module into the chip itself instead of being external, resulting in a smaller size and lower heat generation, while the maximum bandwidth has tripled. The bandwidth of Bluetooth 5.0 reaches 2Mbps, which is twice that of Bluetooth 4.1 supported by the 820A, with the effective transmission distance increased to four times and lower power consumption. [20]

Its advantage lies in its ARM processor architecture, which has mature compatibility with the Android system. Currently, mainstream in-vehicle infotainment systems are mostly based on the Android architecture. This allows the Snapdragon 8155 to achieve rapid adoption due to the chip's compatibility with the system, as well as the openness and richness of the software. [35]

In practical applications, its high computing power supports intelligent features such as full-vehicle voice interaction, Face recognition, gesture recognition, continuous "what you see is what you can say" dialogue, and even voice-controlled parking from outside the vehicle. The chip natively supports virtualization technology, enabling it to run a Type-1 Hypervisor (such as QNX Hypervisor) to create multiple isolated Virtual Machine (VMs) on a single SoC. For

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and navigation. [22]

Key Information

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ZEEKR spent RMB 300 million to upgrade the vehicle computer chip for Zeekr 001 owners free of charge from the Snapdragon 820A to the Qualcomm Snapdragon 8155 Smart Cockpit Computing Platform. Zeekr announced in July that it would provide a free upgrade to the Qualcomm Snapdragon 8155 Smart Cockpit Computing Platform for all Zeekr 001 owners. After the upgrade, the vehicle computer wake-up time was shortened to 2 seconds, cold start speed was increased by 3 times, swipe frame rate was improved by 66.1%, and the average time to open in-vehicle applications was only 1.3 seconds. In early September, Ford announced that it would offer a free upgrade to the Qualcomm Snapdragon SA8155P chip for existing or soon-to-be Ford Mustang Mach-E customers. [14]

Great Wall Motor ' ORA Funky Cat model triggered concentrated complaints from owners because it was advertised as being equipped with the Qualcomm 8155 chip but did not actually use it. [15]

In March 2026, the Toyota Corolla model sparked controversy because the advertised Qualcomm Snapdragon 8155P chip was actually underclocked and had its memory modules replaced, leading to infotainment system lag and slow response from the voice assistant. [39]

Subsequent Development and Impact

The cumulative shipment volume of smart cockpit domain controllers based on the Qualcomm Snapdragon SA8155P platform has exceeded 2 million units. [23]

The chip became a key component in China's new energy vehicle market around 2021, playing a role in promoting the development of vehicle intelligence and bringing attention to cockpit chips as independent hardware. [38]

Related News

On the afternoon of October 14, 2026, Leapmotor released a preview of its pure electric compact car A10s. The Leapmotor A10 is equipped with a dual-chip combination of the SA8155 cockpit chip and the SA8650 ADAS chip. [52]

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